Toward a four-dimensional Yang–Mills construction

Notebook 17: joint fluctuation tails and the one-step Gaussian domain

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Abstract

We prove joint bad-block bounds for correlated Gaussian fluctuations from a counting-coordinate covariance bound. The estimates allow singular observable covariances and overlapping tests. A relative complex quadratic perturbation has a corresponding local estimate without a factor growing with the total volume: untested variables are integrated out before taking absolute values. In four dimensions, an exact fixed-ratio rescaling supplies the required covariance bound for both potential values and first differences. The actual reduced Wilson Hessian at our nonlinear constrained minimizer satisfies the relative quadratic hypothesis, including its constraint-curvature term. A separate extraction of the earlier flat effective-action bounds supplies the current-lattice Gaussian at arbitrary homogeneous averaging depth. These are inputs to the quantum integration and iteration required for the existence argument.

1 Which fluctuation estimate is needed

Our objective is a nontrivial four-dimensional Yang–Mills quantum theory. Notebooks 15 and 16 provide the homogeneous finite-regulator classical minimizer, its ordinary-constraint transfer, local analytic responses, and the minimized action. The next step integrates the fluctuations and sums the regions that leave the analytic chart.

The Gaussian roughness calculation of Notebook 16 distinguishes two different integrations. Integrating every unresolved mode below a fixed macroscopic scale does not give a uniform smooth-field domain. Integrating a fixed-ratio step has different coordinates and different uniform bounds. Both statements must retain the physical pairing: an inverse precision as a physical Hilbert operator is not its raw coordinate covariance matrix.

This notebook proves the joint tail estimate needed for prescribed bad blocks, its complex version, and the fixed-step quadratic identifications supported by our construction. A good-cell estimate alone does not supply a regional nonlinear chart on its complement. The latter construction and the interacting weights are explicit interfaces for the next step.

The proofs have been checked internally by separate derivations and audits. The results are finite-regulator theorems on their stated domains; a continuum existence theorem is the target of their subsequent integration.

2 One fixed-ratio fluctuation step in physical coordinates

We work in dimension four with a fixed odd blocking ratio $L \geq 3$. Fine spacing is $\eta > 0$ and the next retained spacing is $h = L\eta$. The number of retained cells may be arbitrarily large; constants may depend on L and the gauge group, but not on h, η, or compatible periodic volume. This is one blocking step, rather than a simultaneous integration of all fluctuations below a fixed macroscopic scale.

Let $y = x/h$, and write the rescaled potential as $A_s(y) = hA_{\mathrm{phys}}(x)$. Then

$$U(e) = e^{i\eta A_{\mathrm{phys}}(e)} = e^{iA_s(e)/L}, \qquad \nabla_\eta A_{\mathrm{phys}} = h^{-2}\nabla_{1/L} A_s. \tag{1}$$

The four-dimensional coupling-free Wilson action is unchanged by this relabeling: it is the same sum of plaquette densities, each counted once. The physical Maxwell action likewise obeys

$$\eta^4 \sum |d_\eta A_{\mathrm{phys}}|^2 = L^{-4} \sum |d_{1/L} A_s|^2. \tag{2}$$

The corresponding scalar gauge projection and flat averaging constraint are the fixed-L unit-coarse-scale operators of the preceding notebooks after this rescaling.

Write $X_L = \ker Q_L \cap \ker(R_L D_*^T)$, where $D_* = D_{1/L}$, and let $\mathcal{C}_L$ be the Maxwell variational inverse on X_L. For the physical constrained Gaussian Z_{phys} and its rescaling $Z_s = hZ_{\mathrm{phys}}$, the covariance *operators* under the physical pairings and their raw coordinate matrices satisfy

$$\mathcal{C}_h = h^2 \mathcal{C}_L, \qquad \mathrm{Cov}_{\mathrm{count}}(Z_{\mathrm{phys}}) = \eta^{-4} h^2 \mathcal{C}_L, \qquad \mathrm{Cov}_{\mathrm{count}}(Z_s) = L^4 \mathcal{C}_L. \tag{3}$$

The last two are coordinate covariances, rather than operators under weighted physical sums. They follow also by pushing the normalized Gaussian through (2); the constant Lebesgue Jacobian cancels from its normalization.

Define the complete dimensionless value/first-difference vector

$$Y = (\eta Z_{\mathrm{phys}}, \eta^2 \nabla_\eta Z_{\mathrm{phys}}) = (Z_s/L, \nabla_* Z_s/L^2). \tag{4}$$

Equation (3) gives the exact counting covariance

$$\Sigma_Y = S_L \mathcal{C}_L S_L^T, \qquad S_L = (LI, \nabla_*). \tag{5}$$

The previously proved energy sandwiches imply

$$0 \le \Sigma_Y \le K_L I \tag{6}$$

for a fixed finite K_L, uniformly in spacing and volume. Indeed $\mathcal{C}_L \le G_{0,L}$ and $(I, \nabla_*) G_{0,L} (I, \nabla_*)^T$ is uniformly bounded; the extra factor L is fixed. All real trace-orthonormal color components can be included at once.

Every retained cell and a fixed collar contain a fixed number m_L of these scalar tests. A test belongs to at most a fixed number ν_L of such collars. Consequently a covariance-based joint-tail theorem applies with constants independent of the physical scale and the number of cells. Singularity of Σ_Y from gradients or constraints causes no problem for such an upper bound.

2.1 Small nonconstant backgrounds: the actual reduced Hessian

We can also control the quadratic fluctuation form at the nonlinear corrected minimizer from Notebook 15, on its stated small global coarse-data ball. Everything in this subsection is first expressed at retained spacing one, with fine spacing $1/L$. Let

$$A = \xi + H[B - \mathcal{R}(A)], \qquad A_* = A(\xi_*(B), B), \qquad T_A = (I + HD\mathcal{R}(A))^{-1}.$$

Use the physical trace pairing and let K_X be the compression of $K = d^T d$ to X_L. Write $\mathcal{B}_B$ for the Hessian, in the linear ξ coordinate, of the exact reduced Wilson action at $\xi_*(B)$. This is a fine fluctuation Hessian; it is different from the coarse Hessian $D_B^2 V$ of Notebook 16.

For $z, w \in X_L$, put $v = T_{A_*} z$ and $u = T_{A_*} w$. The exact second derivative of the constraint chart gives

$$\mathcal{B}_B[z, w] = \langle v, Ku \rangle + \langle S_{\mathrm{jet}} v, W(A_*) S_{\mathrm{jet}} u \rangle - \langle m(B), D^2 \mathcal{F}(A_*)[v, u] \rangle. \tag{7}$$

Here $W(A_*) = Df_{1/L}(S_{\mathrm{jet}} A_*)$ has norm $O(\|B\|_\infty)$ and $m(B) = O(\|B\|_\infty)$. The last term is retained; a Hessian of the ambient action alone would not be the reduced fluctuation form.

The previously established tangent estimate is

$$\|v\|_2 + \|\nabla_* v\|_2 \le C \langle z, Kz \rangle_{\mathrm{Herm}}^{1/2}, \tag{8}$$

and the same holds for u, w, also on the common complex background ball. To see its complex extension directly, use $v = z - HD\mathcal{R}(A_*)v$, the physical Hilbert smallness $\|D\mathcal{R}(A_*)\|_{2\to 2} = O(\|B\|_\infty)$, the bounded $H, \nabla_* H$, and the flat coercive bound on X_L.

The flat mixed terms vanish, giving the exact bilinear identity

$$\langle v, Ku \rangle - \langle z, Kw \rangle = \langle D\mathcal{R}(A_*)v, qD\mathcal{R}(A_*)u \rangle. \tag{9}$$

The right side is $O(\|B\|_\infty^2)$ relative to the two Maxwell energy norms. The middle term of (7) is $O(\|B\|_\infty)$ by (8). The genuine two-Hilbert-direction constraint estimate bounds the last term by the same order. Consequently, on a fixed smaller complex ball,

$$\mathcal{B}_B = K_X^{1/2}(I + E_B)K_X^{1/2}, \qquad E_B^T = E_B, \qquad \|E_B\|_{2\to 2} \le C \|B\|_\infty . \tag{10}$$

The normalized matrix E_B is holomorphic in B. The square root is that of the fixed real positive K_X; the norm is the Hermitian operator norm, while the displayed transpose is the analytic bilinear transpose.

For real B, choose the radius so that $\|E_B\| \le \epsilon < 1$. The centered Gaussian $\zeta_s \in X_L$ for this reduced quadratic form has covariance operator $\mathcal{C}_B = J_X \mathcal{B}_B^{-1} J_X^T$ and satisfies

$$(1+\epsilon)^{-1}\mathcal{C}_L \le \mathcal{C}_B \le (1-\epsilon)^{-1}\mathcal{C}_L. \tag{11}$$

Its dimensionless *chart-coordinate* vector $(\zeta_s/L, \nabla_* \zeta_s/L^2)$ therefore has counting covariance bounded by $(1-\epsilon)^{-1} K_L I$.

The linearized physical-potential fluctuation is $T_{A_*}\zeta_s$. It also has a uniformly bounded dimensionless value/gradient covariance. Indeed (8) bounds $S_L T_{A_*} J_X K_X^{-1/2}$ uniformly, so

$$\left\| S_L T_{A_*} \mathcal{C}_B T_{A_*}^T S_L^T \right\|_{2\to 2} \le C_L (1-\epsilon)^{-1}. \tag{12}$$

Here the finite-dimensional maps on X_L are read with the same inclusions as in (11). This is a real Gaussian statement at a real background.

For complex B, (10) instead defines a normalized complex Gaussian functional with strictly positive Hermitian real part of its precision. Estimates for fixed real coordinate tests can be obtained from its finite-dimensional marginals. A cutoff whose defining tests themselves depend on complex B is a different problem; no holomorphy is inferred by placing an indicator around such moving complex tests.

2.2 The local domain for the nonlinear chart

At the fixed mesh $1/L$, bounded values control all physical first differences and their Hölder seminorms with constants depending only on L and α. Thus on the event that *all fine value coordinates in the regulator* of the dimensionless chart vector have modulus at most R,

$$\|g\zeta_s\|_{N_\alpha} \le C_{L,\alpha} g R. \tag{13}$$

Choose gR small enough to stay inside the nonlinear chart and its analytic action domain. If $R(g) = (\log(1/g))^p$ for fixed $p > 0$, then both $gR(g)$ and $gR(g)^3$ tend to zero. This is compatible with a large Gaussian-tail threshold. The same norm comparison on a fixed cell and its stencil collar is a local comparison of fields. It does not extend the globally defined nonlinear minimizer chart through an arbitrary complement of bad cells; that requires a regional construction with its own boundary estimates. It is the relevant fixed-ratio local scaling, rather than a uniform smooth bound on all unresolved macroscopic modes.

On arbitrarily large volumes the event that *every* cell is good need not have large probability. The quantum step requires joint bounds on prescribed bad cells and a summation over their connected regions. The covariance bound supplies the first of these ingredients. Integrating the nonlinear remainder, the Haar and coordinate Jacobians, and bad regions remains a further task; the present Gaussian comparison does not replace those integrals.

3 Joint Gaussian bounds for prescribed bad blocks

The input to this section is a counting-coordinate covariance operator bound. A bound on each coordinate variance alone does not suffice. We first give a real Gaussian theorem, including degenerate covariances and overlapping tested sets. We then prove a normalized complex Gaussian extension under an explicit relative quadratic hypothesis. All statements are initially finite-dimensional; their constants do not depend on the ambient dimension or the number of tested blocks.

3.1 The covariance input and microscopic blocks

Let $Y = (Y_i)_{i \in I}$ be a centered real Gaussian vector with counting-coordinate covariance

$$0 \le \Sigma \le KI, \qquad K > 0. \tag{14}$$

For each block b let $I_b \subset I$ be a nonempty tested set. Assume fixed constants m, ν satisfy

$$|I_b| \le m, \qquad \sup_i \#\{b : i \in I_b\} \le \nu. \tag{15}$$

Thus m is the number of scalar tests per block and ν is their overlap multiplicity. Define

$$\mathsf{B}_b(R) = \{\max_{i \in I_b} |Y_i| > R\}, \qquad R > 0. \tag{16}$$

Tests may be strongly correlated. The overlap counts repeated Y coordinates, not overlap of the underlying field variables from which a derivative test was formed. Enlarging a block to include a fixed stencil changes m and ν by fixed factors.

Theorem 3.1 (Joint real Gaussian bad-block bound)**.** *For every finite set S of k blocks,*

$$\mathbb{P}\left(\bigcap_{b \in S} \mathsf{B}_b(R)\right) \le \left[m\sqrt{2}\exp\left(-\frac{R^2}{4\nu K}\right)\right]^k. \tag{17}$$

The prefactor $m\sqrt{2}$ may instead be $2^{m/2}$, and the right side may be replaced by its minimum with one.

Proof. Choose one coordinate $i_b \in I_b$ for each block. The vector $W = (Y_{i_b})_{b \in S}$ has covariance at most νKI: the coordinate-selection map, with repetitions allowed, has squared operator norm at most ν. Diagonalizing this nonnegative covariance, allowing zero eigenvalues, gives

$$\mathbb{E}e^{t|W|^2} = \det(I - 2t\Sigma_W)^{-1/2} \le (1 - 2t\nu K)^{-k/2}, \qquad 0 \le t < (2\nu K)^{-1}. \tag{18}$$

If every chosen coordinate exceeds R in absolute value, $|W|^2 > kR^2$. Take $t = (4\nu K)^{-1}$ in Markov's inequality. The resulting bound is $2^{k/2}e^{-kR^2/(4\nu K)}$. There are at most m^k coordinate choices, and their union contains the event in (17).

Alternatively collect every tested coordinate in the selected blocks, retaining repetitions. This vector has at most mk entries and covariance at most $\nu K I$. On the all-bad event its squared norm is at least kR^2. Applying (18) with dimension mk proves the alternative prefactor. Neither proof assumes independent blocks. □

If a local norm obeys $N_b(Y) \le c_b \max_{i\in I_b} |Y_i|$, its bad event is bounded by the theorem with threshold R/c_b. For a Euclidean block norm, the second proof applies directly to $\sum_{i\in I_b} Y_i^2 > R^2$, with prefactor $2^{m/2}$. For gY replace R by R/g, equivalently K by g^2K.

3.2 Exponential squares, entropy, and connected sets

Put $E_b(Y) = \sum_{i\in I_b} Y_i^2$. For nonnegative t_b of finite support define

$$a_i = \sum_{b: i\in I_b} t_b, \qquad A = \operatorname{diag}(a_i).$$

Then $\|A\| \le \nu t_{\max}$ and $\operatorname{Tr} A \le m \sum_b t_b$. The exact Gaussian determinant, applied to $\Sigma^{1/2} A \Sigma^{1/2}$, gives

$$\boxed{\log \mathbb{E} \exp\left(\sum_b t_b E_b(Y)\right) \le \frac{Km \sum_b t_b}{1 - 2K\nu t_{\max}}, \qquad 2K\nu t_{\max} < 1.} \tag{19}$$

Indeed each eigenvalue is at most $K\nu t_{\max}$, their sum is at most $K \operatorname{Tr} A$, and $-\log(1-x) \le x/(1-x_{\max})$ for $0 \le x \le x_{\max} < 1$. This proof also permits singular covariance.

For a probability law ρ absolutely continuous with respect to the Gaussian law μ, the entropy variational inequality therefore gives

$$\sum_b t_b \mathbb{E}_\rho E_b(Y) \le D(\rho\|\mu) + \frac{Km \sum_b t_b}{1 - 2K\nu t_{\max}}. \tag{20}$$

Truncation gives the same assertion with extended expectations when necessary.

Let $p(R)$ be the bracket in (17), decreased to at most one if desired. Expanding a finite product with $z_b \ge 0$ yields

$$\mathbb{E} \prod_b (1 + z_b \mathbf{1}_{\mathsf{B}_b(R)}) \le \prod_b (1 + p(R) z_b). \tag{21}$$

In particular, for N tested blocks and $s \ge 0$, the exponential moment of their bad-block count is at most $(1 + p(R)(e^s - 1))^N$. This is a moment statement, not an assertion that the process is independent or stochastically dominated for every increasing event.

Give the blocks an adjacency graph of maximum degree at most $\Delta \ge 1$. The number of connected k-block sets containing a specified root is at most $(4\Delta)^{k-1}$. One way to see this is to choose a canonical rooted traversal tree: ordered rooted tree shapes cost at most 4^{k-1}, and each edge has at most Δ neighbor choices. Overcounting only enlarges the bound. Consequently, whenever $4\Delta e^a p(R) < 1$,

$$\boxed{\mathbb{E} \sum_{\substack{S \ni b_0 \\ S \text{ connected}}} e^{a|S|} \prod_{b\in S} \mathbf{1}_{\mathsf{B}_b(R)} \le \frac{e^a p(R)}{1 - 4\Delta e^a p(R)}.} \tag{22}$$

This is a summable Gaussian bad-set estimate. It does not estimate an interacting polymer activity or prove its strict background dependence.

3.3 The homogeneous flat applications

For the previously constructed d4 constrained flat Gaussian, the covariance under the physical pairing is $\mathcal{C}$, whereas its coordinate covariance is $\mathrm{Cov}_{\mathrm{count}}(Z) = \eta^{-4}\mathcal{C}$. The dimensionless derivative vector therefore has exactly

$$\mathrm{Cov}_{\mathrm{count}}(\eta^2 \nabla Z) = \nabla \mathcal{C} \nabla^T \le K_D I. \tag{23}$$

The inequality follows from the proved real order $\mathcal{C} \le \Gamma \le G_0$ and the uniform first-derivative sandwich of G_0. Input and output fine spaces have the same physical scalar measure, so the operator norm of this coordinate matrix is unchanged on passing to counting Hilbert norms. The color basis is real trace-orthonormal.

Fixed microscopic blocks contain a fixed number of tests, so Theorem 3.1 and (19)–(22) apply uniformly over the full homogeneous refinement. Rescaled curls are fixed local combinations of this derivative vector and have the same conclusion with fixed norm and incidence factors. The sampled microscopic Hölder term satisfies $\eta^{2+\alpha}[\nabla Z]_\alpha \le C \left\|\eta^2 \nabla Z\right\|_\infty$ and can be included at fixed cost. A unit physical cell contains M^4 fine sites, so treating it as one block here would make m grow with M.

For the full macroscopic Gaussian the potential vector instead has covariance

$$\mathrm{Cov}_{\mathrm{count}}(\eta Z) = \eta^{-2}\mathcal{C}. \tag{24}$$

Its counting-operator norm is not uniformly bounded over refinement. For $M \ge 3$, at zero coarse momentum a transverse fine alias $r = e_1$, polarized in direction 2, lies in the constraints and has Maxwell eigenvalue $4M^2 \sin^2(\pi/M) \le 4\pi^2$. Thus (24) has norm at least $M^2/(4\pi^2)$. Pointwise potential variance bounds do not remove this operator obstruction.

For the first bare homogeneous step of a fixed ratio L, the exact physical scaling established in the preceding section instead gives the joint value/derivative covariance

$$\Sigma_{\mathrm{step}} = S_L \mathcal{C}_L S_L^T \le K_L I, \qquad S_L = \begin{pmatrix} LI \\ \nabla_{1/L} \end{pmatrix}. \tag{25}$$

Here L is fixed and the number of retained cells is arbitrary. The same theorem therefore applies to both components together, including their cross covariances. The real reduced-Hessian comparison of that bare step enlarges K_L by at most $(1-\epsilon)^{-1}$ for chart-coordinate tests. At real backgrounds it also gives a uniform bound for the separately identified linearized-potential tests, so the real theorem applies to those as well.

These applications do not replace the precision in a later effective step by the bare fixed-L Maxwell form. A later coarse precision and its separate current terms require their own covariance identification and uniform bound.

3.4 A normalized complex quadratic extension

Whiten the real baseline Gaussian on its nondegenerate slice: let ζ be standard real Gaussian in $\mathbb{R}^N$ and let $Y = A\zeta$, with $AA^T \le KI$. The observable covariance may be singular. Suppose E is complex symmetric and

$$E^T = E, \qquad \|E\|_{2\to 2} \le \epsilon \le \frac{1}{4}. \tag{26}$$

Define the normalized complex Gaussian functional

$$d\mu_E(\zeta) = \det(I+E)^{1/2} e^{-\zeta^T E \zeta/2}\, d\gamma_N(\zeta), \tag{27}$$

with the analytic determinant branch equal to one at $E = 0$. Its total mass is one. For a physical quadratic perturbation V_X of the positive baseline precision K_X, condition (26) means exactly

$\left\|K_X^{-1/2} V_X K_X^{-1/2}\right\| \le \epsilon$. An absolute smallness assertion about matrix entries is not a substitute for this relative bound. At complex E the functional is not a positive probability law.

Theorem 3.2 (Complex Gaussian local-event bound)**.** *Under* (15) *and* (26)*, every set S of k blocks satisfies*

$$\boxed{\left|\mu_E\left(\bigcap_{b\in S} \mathsf{B}_b(R)\right)\right| \le \left[2^m \exp\left(-\frac{R^2}{6\nu K}\right)\right]^k.} \tag{28}$$

There is no ambient-N factor. The events use fixed real linear tests $Y = A\zeta$ and fixed real thresholds.

Proof. All tested coordinates in S span a real row space of dimension $r \le mk$. Make a real orthogonal change of the whitened coordinates so that these tests depend only on $x \in \mathbb{R}^r$; denote the untested orthogonal coordinates by z. Integrating z exactly gives a normalized complex Gaussian marginal on x with precision

$$B = I + E_{11} - E_{12}(I + E_{22})^{-1} E_{21}. \tag{29}$$

Therefore

$$\|B - I\| \le \epsilon + \frac{\epsilon^2}{1-\epsilon} = \frac{\epsilon}{1-\epsilon} =: s \le \frac{1}{3}, \qquad \operatorname{Re} B \ge (1-s)I =: \beta I, \qquad \beta \ge \frac{2}{3}. \tag{30}$$

The real-part assertion is an inequality for the real quadratic form, equivalently the Hermitian numerical range of the complex symmetric matrix. Since $|\det B| \le \|B\|^r \le (1+s)^r$, the modulus of the normalized marginal density is bounded by

$$\left(\frac{1+s}{\beta}\right)^{r/2} d\gamma_{\beta^{-1} I_r}(x) \le 2^{r/2} d\gamma_{\beta^{-1} I_r}(x). \tag{31}$$

Under this dominating real Gaussian the tested covariance norm is at most K/β. Collect all tests in the selected blocks, including repetitions. Its dimension is at most mk, overlap at most ν, and covariance norm at most $\nu K/\beta$. On the all-bad event its squared norm exceeds kR^2. Equation (18), with $t = \beta/(4\nu K)$, bounds its probability by $2^{mk/2} \exp[-\beta k R^2/(4\nu K)]$. Multiply by $2^{r/2} \le 2^{mk/2}$ and use $\beta \ge 2/3$. This proves (28), including degenerate observable covariance and the case $r = 0$. □

The proof also bounds a bounded measurable factor depending only on the same tested marginal, multiplied by the all-bad indicator, by its supremum times the right side of (28). It does not bound the integral of the modulus of the full N-dimensional density over that event without an N-dependent factor. Untested variables must be integrated out before taking the modulus. An observable using further coordinates requires enlarging the marginal and accounting for its dimension.

For fixed real-coordinate events, the normalized integral is holomorphic in E on the complex operator ball. At each finite N, the real-part lower bound gives Gaussian domination on compact subballs and the determinant branch is analytic. If E depends holomorphically on sources in a known neighborhood remaining inside (26), Cauchy on that neighborhood gives factorial source derivatives of these local event integrals, retaining the displayed bad-block decay on a slightly larger parameter domain. This requires a proved relative perturbation domain; it does not produce one for an unmatched source Hessian.

In particular an indicator around tests that themselves move with a complex background is not covered by this holomorphy argument. For the reduced first-step Hessian, the fixed real chart-coordinate tests are admissible. Background-dependent linearized-potential tests are covered by the real covariance theorem at real backgrounds, but require a separate analytic cutoff construction at complex backgrounds.

Expansion of a finite product also gives

$$\left|\mu_E \prod_b (1 + z_b \mathbf{1}_{\mathsf{B}_b(R)})\right| \le \prod_b (1 + p_{\mathrm{c}}(R) z_b), \qquad p_{\mathrm{c}}(R) = 2^m e^{-R^2/(6\nu K)}, \qquad z_b \ge 0. \tag{32}$$

One must not cap p_{c} at one merely because the total mass is normalized: complex expectations of events are not probabilities. The same connected-set counting as (22) bounds the sum of absolute local event expectations whenever its entropy inequality holds.

3.5 Source match and application boundary

For the homogeneous flat derivative application, the action, coupling and physical pairing are the deposited conventions at Raw I:464–479, I:12756–12760 and I:15643–15648. The flat corrected average and its homogeneous composition are matched at I:24349–24373 and I:47046–47048; the scalar projection is matched at I:23937–23944. The analytic input actually used is the established variational $\mathcal{C}$ and full-G_0 first-derivative sandwich. These are checks against the pinned deposited transcription, not a new original-paper verification.

The real theorem therefore applies to that explicit homogeneous flat constrained Gaussian, and to the fixed-ratio first bare step with the covariance identification already proved. The complex theorem applies under its explicit relative quadratic hypothesis; for the reduced first bare step this hypothesis is supplied by the preceding Hessian calculation after shrinking the background ball. Later effective current-dependent steps require their own identification.

These joint bounds are stronger than marginal variance or a single union bound. They still do not integrate nonlinear interactions, Haar or chart Jacobians, or regions outside the chart. They do not supply a complete retained-history covariance, an interacting polymer expansion, or a quantum continuum limit.

4 The Gaussian at an arbitrary current renormalization scale

The fixed-ratio bare Gaussian and the Gaussian used after earlier classical minimizations are different objects. We now extract the uniform bounds for the latter. They are already contained in Notebooks 05–07; this section supplies the current-scale probability normalization and records how Notebook 15 removes the earlier homogeneous smooth-minimizer hypothesis.

4.1 The retained coordinates and their exact precision

Set the current lattice spacing to one. The original spacing is M^{-1}, with $M = L^k$, while the next blocking ratio $L \ge 3$ is fixed and odd. All regulators are compatible periodic tori. Let

$$q_M(B) = \inf_{Q_M^{\mathrm{st}} A = B} \left\| d_{1/M} A \right\|_{1/M}^2$$

be the classical flat effective quadratic form. The fine norm has volume factor M^{-d} and the current norm is counting measure. Next-step maps are denoted by $Q_L^\flat, T_L, K_L$: respectively the rooted constraint, averaged root-to-site contour map, and pivot insertion. These are the fixed-L maps of Notebook 05. The map K_L inserts a field in $\ker Q_L^\flat$ and is the identity on the retained nonpivot coordinates. It is not a gauge-selected fine minimizing extension.

The actual flat exponential-gauge quadratic model on the retained variables z is

$$A_{M,0} = K_L^T (q_M + T_L^T T_L) K_L, \qquad d\mu_{M,0}(z) = (2\pi)^{-n/2} \det(A_{M,0})^{1/2} e^{-z^T A_{M,0} z/2}\, dz. \tag{33}$$

Here n counts independent real bond/color coordinates once, using a real color basis orthonormal for the normalized trace. Thus

$$\mathrm{Cov}_{\mathrm{count}}(z) = A_{M,0}^{-1}. \tag{34}$$

There is no factor M^d in (34): these are *current-unit* coordinates, not original fine potential values. The coupling has already been removed from the Gaussian by the change of field $B = g_k K_L z$ at linear order. The nonlinear constraint correction, Haar density and coordinate Jacobian are kept outside the chosen Gaussian in (33). In particular, a coupling-suppressed quadratic Haar term may be part of the interaction; no vanishing of that term is asserted.

4.2 Uniform spectral bounds, including all previous depths

The normalized cochain contraction in Notebook 06 gives

$$\|d_1 B\|_1^2 \le q_M(B) \le C_d \, \|B\|_1^2 , \qquad C_d = 2^{d+6}(d-1). \tag{35}$$

The compact lift proves the upper bound for odd $M \ge 3$; $M = 1$ uses the identity lift and satisfies the same displayed bound. These estimates hold for all current fields, not just a transverse Fourier sector.

For clarity we record the fixed geometric constants of Notebook 06. On the cube of 3^d neighboring L-blocks form the finite matrix

$$\mathcal{M}_L = d_\Omega^T d_\Omega + T_{0,\Omega}^T T_{0,\Omega} + Q_{\mathrm{st},\Omega}^T Q_{\mathrm{st},\Omega}, \quad \sigma(d,L) = \lambda_{\min}(\mathcal{M}_L) > 0.$$

Its kernel is trivial: zero curl gives a gradient, the tree rows make its scalar potential constant on each block, and the averaging rows equate neighboring block constants. Summing the local inequality gives the axial gap $\gamma_{\mathrm{loc}} = \sigma(d,L)/3^d$ on $\ker Q_L^{\mathrm{st}} \cap \ker T_0$. Periodic repetitions are counted with multiplicity, so the same constant applies on short compatible tori.

Put

$$\begin{aligned} &d_0 = 2\sqrt{d}, \qquad t_0 = \sqrt{d(L-1)(L^d-1)}, \qquad r_L = 1 + 2d(L-1)/L, \\ &K_*^2 = 1 + L^{2d}(2d+2)r_L^2, \\ &\gamma_* = \left((1+d_0 t_0)^2/\gamma_{\mathrm{loc}} + d_0^2\right)^{-1}, \qquad B_* = (C_d + t_0^2)K_*^2. \end{aligned} \tag{36}$$

The flat gauge bridge of Notebook 05 gives

$$q_M(B) + \|T_L B\|^2 \ge \gamma_* \, \|B\|^2 \quad (Q_L^\flat B = 0).$$

One can see the constraint compatibility directly. First subtract the rooted gradient $D(T_L B)$ to obtain B_T with $T_L B_T = Q_L^{\mathrm{st}} B_T = 0$. Convert it by a block-mean-zero gradient to the tree representative B_0. Their curls agree and $\|B_T\| \le (1 + d_0 t_0)\, \|B_0\|$. Apply the axial gap and weighted Cauchy–Schwarz to $B = B_T + D(T_L B)$. This argument uses the lower inequality in (35); it does not confuse the rooted and straight constraints.

Since z is a coordinate restriction of $K_L z$ and $\|K_L\| \le K_*$, the full precision sandwich is

$$\gamma_* I \le A_{M,0} \le B_* I, \qquad B_*^{-1} I \le A_{M,0}^{-1} \le \gamma_*^{-1} I. \tag{37}$$

All constants are independent of k, M and the torus volume. There is no need to replace this proved bound by an assumed transverse ellipticity statement.

4.3 Weighted locality and its scope

Group the retained coordinates by their fixed next-step L-cells. For a block matrix F, use the weighted Schur norm

$$\|F\|_\kappa = \max\left\{\sup_i \sum_j e^{\kappa|i-j|_1}\,\|F_{ij}\|\,, \sup_j \sum_i e^{\kappa|i-j|_1}\,\|F_{ij}\|\right\},$$

with periodic distance on finite regulators. Notebook 07, its flat-decay theorem and section on periodic gluing and the reduced inverse, prove constants $\kappa_* > 0, C_* < \infty$, depending only on fixed geometry, such that

$$\|A_{M,0}\|_{\kappa_*} + \left\|A_{M,0}^{-1}\right\|_{\kappa_*} \le C_*. \tag{38}$$

The proof there first establishes a uniform holomorphic Fourier strip for q_M, then uses fixed-range K_L, T_L, the real gap (37), and a smaller strip for the inverse. Exact periodization gives the same conclusion on compatible tori. This additional locality does not follow from a spectral sandwich alone. It concerns the flat retained-coordinate matrix; it is not strict locality in a varying background.

4.4 Joint value-and-difference tails on the current lattice

Let ∇_1 contain all forward unit differences of every bond component, and define

$$Y = \begin{pmatrix} K_L z \\ \nabla_1 K_L z \end{pmatrix}. \tag{39}$$

The elementary bound $\|\nabla_1\| \le 2\sqrt{d}$ and (37) imply

$$0 \le \mathrm{Cov}_{\mathrm{count}}(Y) \le K_{\mathrm{cur}} I, \qquad K_{\mathrm{cur}} = (1+4d)K_*^2/\gamma_*. \tag{40}$$

Because K_L and the difference stencil have fixed range, (38) also supplies exponential block decay for this tested covariance, with a fixed loss in its constant.

If each tested cell uses at most m scalar coordinates and every coordinate is tested in at most ν cells, the real joint-tail theorem 3.1 gives

$$\mathbb{P}\left(\max_{i\in I_b}|Y_i| > R \text{ for every } b \in S\right) \le \left[m\sqrt{2}\exp\left\{-\frac{R^2}{4\nu K_{\mathrm{cur}}}\right\}\right]^{|S|}. \tag{41}$$

Fixed current-scale cells, including fixed collars for derivatives, have m, ν independent of M. The Gaussian is correlated; independence is neither asserted nor needed. For $g_k Y$, replace R by R/g_k. If $R(g) = c(\log g^{-2})^p$ with $p > 1/2$, the bracket at radius $R(g)$ is smaller than any fixed power of g below a sufficiently small ceiling.

To restore the physical current spacing a, write $Z_{\mathrm{phys}}(x) = a^{-1}(K_L z)(x/a)$ in four dimensions. Then

$$(aZ_{\mathrm{phys}}, a^2\nabla_a Z_{\mathrm{phys}}) = Y.$$

This is the current-step normalization. It differs from treating all original fine variables between M^{-1} and one as a single fluctuation. It also differs from the bare fixed-ratio formula $(LI, \nabla_{1/L})C_L(LI, \nabla_{1/L})^T$, whose coordinates use next-coarse spacing one. Both statements are consistent after their scales and Gaussian models are specified.

4.5 What is now identified with the deposited source

Notebook 07's source section records the chain from the deposited small-field step to (33): Raw II:73865–73950 straightens the constraint, rescales by g_k, and eliminates the pivots; Raw I:46960–47000 fixes the pivot coefficient convention. At the trivial field the action Hessian is q_M, the exponential gauge Hessian is $T_L^T T_L$, and the action gradient vanishes. Consequently the quadratic start of the constraint-straightening correction does not change this Hessian. Haar and Jacobian factors remain in the interaction.

The smooth-minimizer premise in that earlier identification is now supplied, in the homogeneous setting, by Notebook 15's theorem on transfer to the ordinary homogeneous averaging constraint. It constructs an analytic local branch for the ordinary iterated group average, with real action minimality in a fixed small global potential class. At each finite regulator the ordinary constraint is an analytic submersion at the identity. Its derivative is onto, as follows from the fixed-step pivot right inverses and their composition. Thus every tangent-feasible field is the tangent to a feasible curve. Comparing the local minimum with these curves shows that its second derivative is the quadratic variational infimum for the rooted derivative. Notebook 06's explicit gradient adjustment identifies that infimum with the displayed Q_M^{st} infimum. The gradient adjustment preserves energy; it does not identify the two averaging maps as maps on potentials.

Accordingly (33) is the homogeneous flat source Gaussian under the deposited ordinary-average and normalization conventions already matched in those notebooks. The nonlinear branch need not equal an independently chosen axial representative: the locally minimizing action and its flat Hessian agree modulo gauge. This does not re-certify every original journal convention, extend the result to mixed histories or holes, or identify a nonflat independent-current analytic family.

4.6 A distinct fixed-slice perturbation consequence

There is also a useful quadratic model without pivot coordinates. Let $X_L = \ker Q_L^{\mathrm{st}} \cap \ker(R_L D_1^T)$ be the fixed flat source-gauge slice for the unit-to-L step, and let J_L be its isometric inclusion. The fixed-L Maxwell covariance proved in Notebooks 09 and 12 implies

$$J_L^T d_1^T d_1 J_L \geq c_L I$$

for some $c_L > 0$, uniformly in compatible volume. For example one may take $c_L = (L^2 \|C_L\|)^{-1}$, where C_L uses next-coarse spacing one. Hence

$$c_L I \leq J_L^T q_M J_L \leq C_d I. \tag{42}$$

This gives the same kind of current-scale value-and-difference tail bound on X_L, uniformly in M.

Notebook 16 proves that the ordinary coarse classical Hessian satisfies $D^2 V_M(B) = q_M + O(\|B\|_\infty)$ in weighted row and column norms. Its compression to this fixed slice therefore has a relative perturbation bounded by $C\|B\|_\infty / c_L$. This is a valid input to the abstract complex relative-quadratic tail theorem on a sufficiently small common background ball. It is an effective classical Gaussian family in a fixed linear slice. It is not automatically the source's nonlinear fiber Gaussian: the latter also contains chart-curvature/current terms, exponential gauge fixing, and an exact cutoff-coordinate prescription. Notebook 19 now performs that additional physical-branch identification, keeping both chart-curvature terms and the actual absolute gauge function. Nor is weighted locality of this compressed inverse inferred by conjugating with weights which do not preserve X_L; the weighted inverse assertion (38) concerns the separately proved pivot-coordinate family.

5 The next integration and iteration

For a logarithmic cutoff $R(g) = (\log(1/g))^p$ with $p > 1/2$, the Gaussian bad-block factor is smaller than any fixed power of g, while $gR(g)^3$ tends to zero. These two estimates are compatible: the same cutoff can suppress bad blocks and make analytic cubic action remainders small. Jacobian terms that begin linearly require their own local bounds; they are not cubic merely because the action remainder is cubic.

The quantum step requires independent fixed-dimensional integration coordinates, their normalized covariance, and localized estimates for the complete interaction. The flat pivot covariance above supplies such coordinates on the homogeneous regulator. The singular value/gradient vector is useful for tests, but is not an independent cell coordinate system for a Gaussian density expansion. For complex continuation the tests used here remain fixed on the real integration variables.

Notebook 18 gives a convergent cutoff expansion with overlapping interaction supports under an explicit weighted coefficient bound. Notebook 19 identifies the nonflat physical Gaussian and bounds the local straightening, Haar and gauge factors. Notebook 20 verifies the complete coefficient and precision hypotheses for the bare Wilson small-field step, with all normalization terms. Notebook 21 supplies strict localization of the minimized classical action uniformly in earlier depth and closes one integration of that reference action. Later quantum steps require transport to the next coarse background and control of the accumulated interactions. Bad regions require regional constructions and their normalization estimates. Iterating those bounds must also control the running couplings and preserve the observable hierarchy needed by the continuum construction.

The output sought from the iteration remains a common nontrivial gauge-invariant continuum hierarchy with the Euclidean covariance, positivity, regularity and Yang–Mills identification specified in Notebook 01. The gap argument is a further target after that existence construction is closed.

References

[1] *Toward a four-dimensional Yang–Mills construction*, research Notebooks 5–16 in this workspace, 16 October 2026. The flat reduced covariance, constrained Maxwell inverse, nonlinear minimum and classical response theorems are used here.

[2] E. Sussman and collaborators, Yang–Mills-LLM-experiment, deposited raw accounts at commit `e392ef45013a581664bab76b90867170b30fe10e`. The Yang–Mills files are unchanged at checked commit `785a68fdeef78a1da854c4807fe1f17b597d05e7`.

[3] T. Bałaban, *Renormalization group approach to lattice gauge field theories I*, Commun. Math. Phys. **109** (1987), 249–301, original article copy. The source-coordinate match is discussed explicitly above.